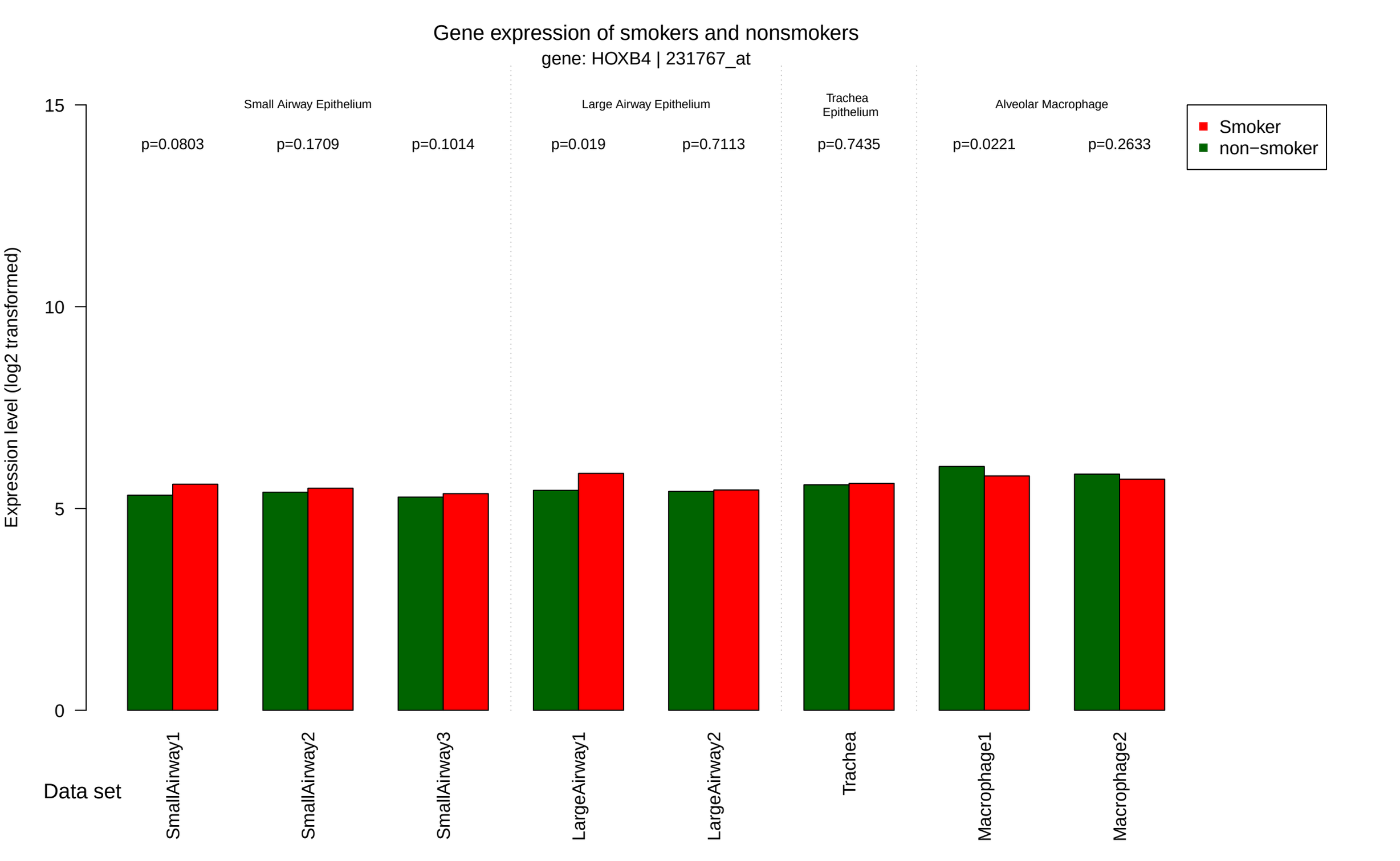
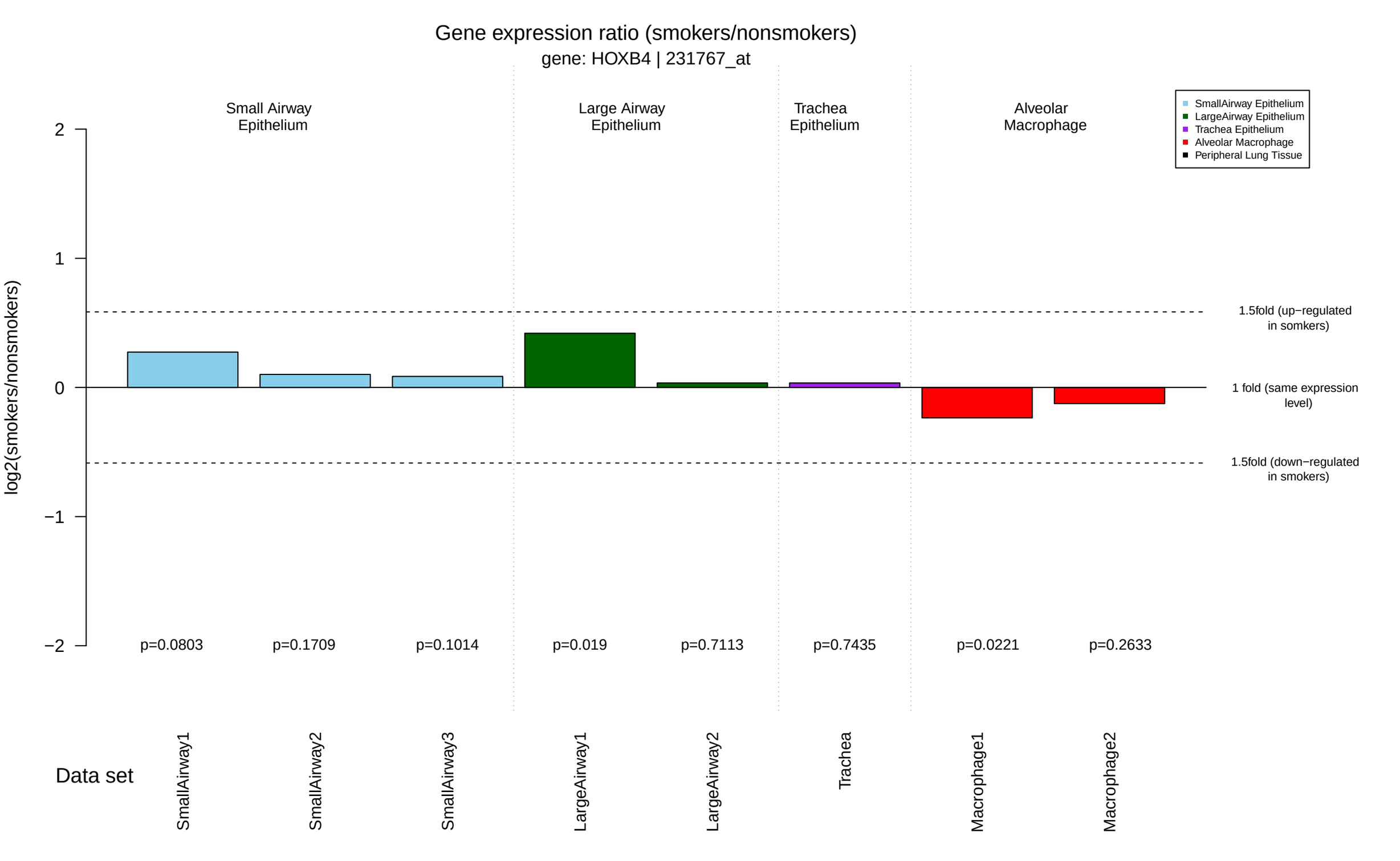
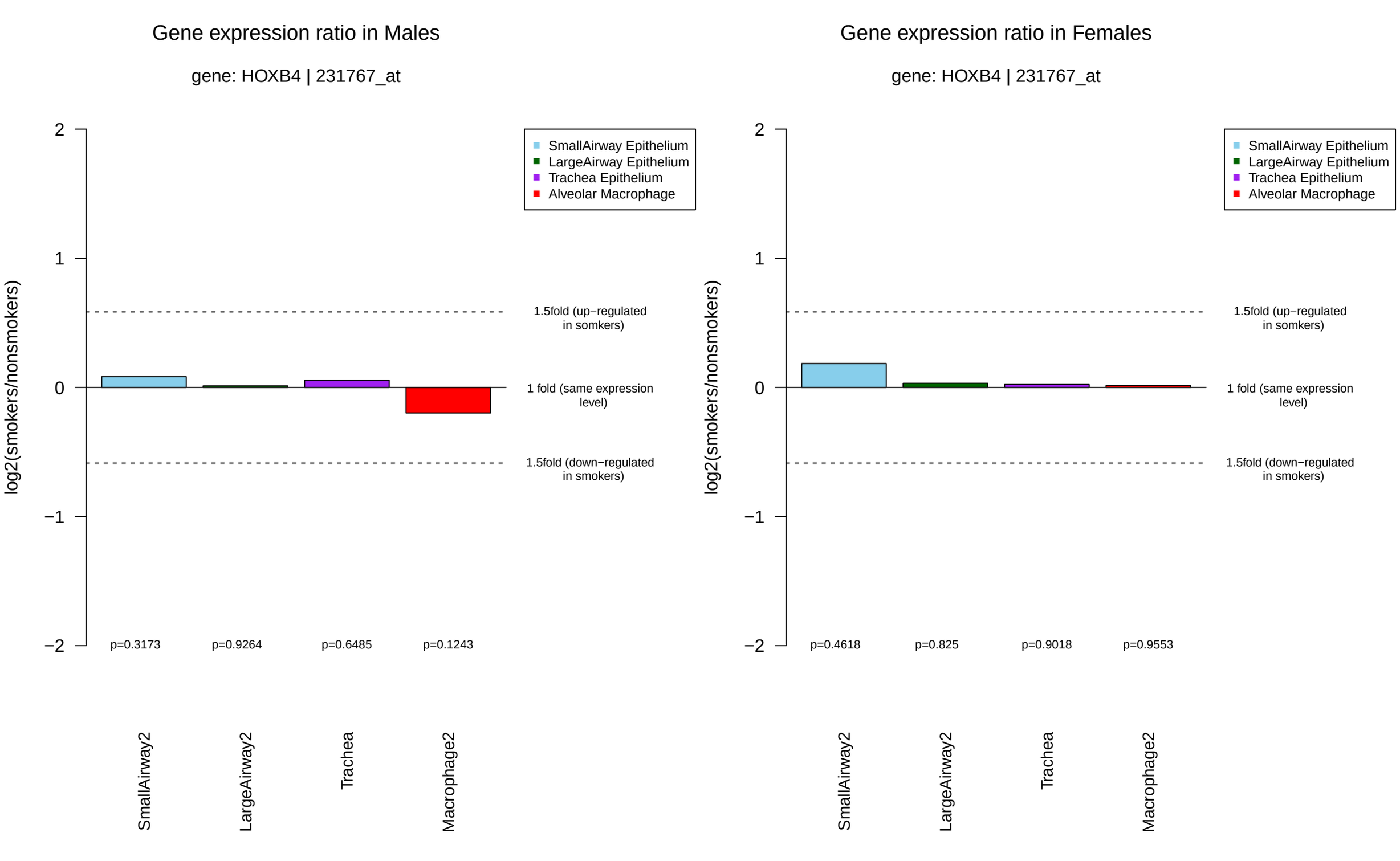
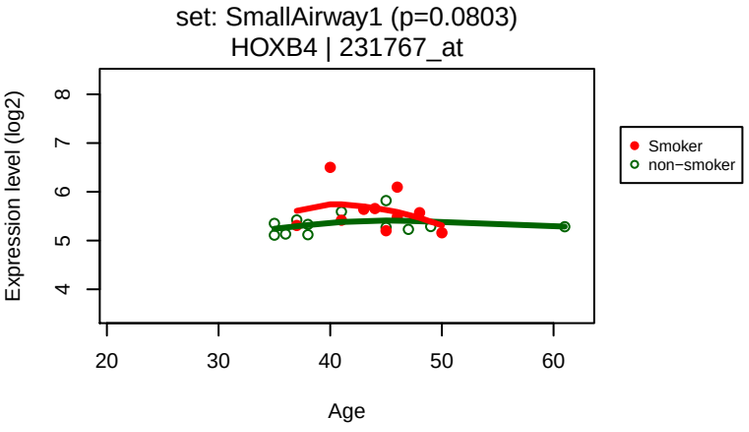
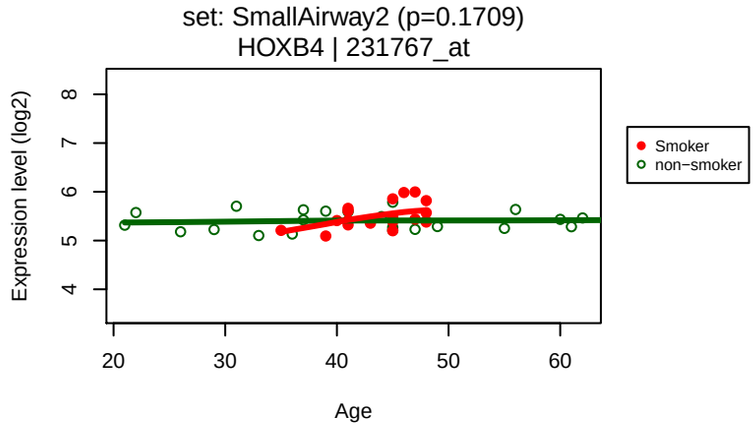
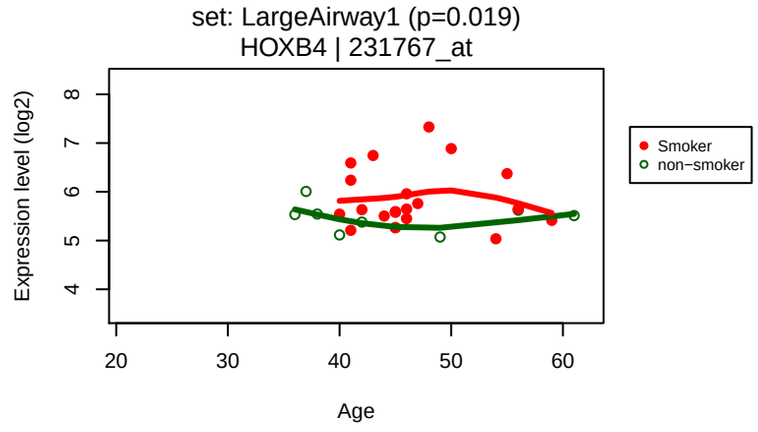

Gene expression of smokers and nonsmokers
gene: HOXB4 | 231767_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0803
p=0.1709
p=0.1014
p=0.019
p=0.7113
p=0.7435
p=0.0221
p=0.2633
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

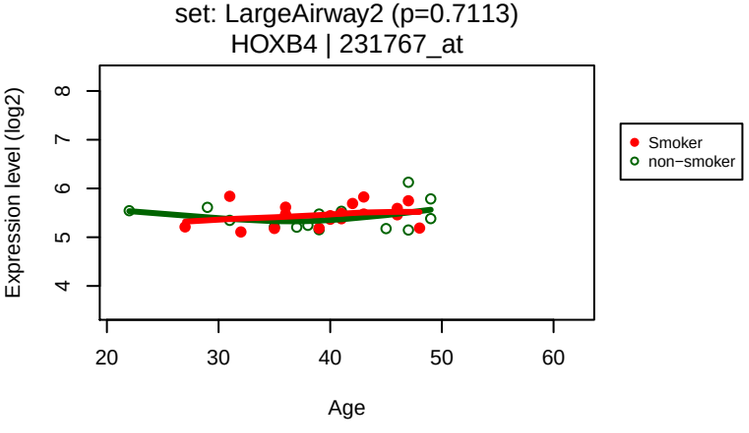

Gene expression ratio (smokers/nonsmokers)
gene: HOXB4 | 231767_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0803
p=0.1709
p=0.1014
p=0.019
p=0.7113
p=0.7435
p=0.0221
p=0.2633
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

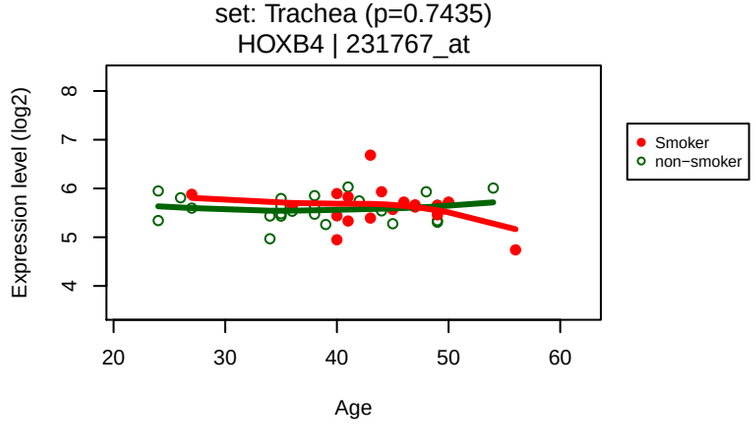

Gene expression ratio in Males
gene: HOXB4 | 231767_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3173
p=0.9264
p=0.6485
p=0.1243
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HOXB4 | 231767_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4618
p=0.825
p=0.9018
p=0.9553
SmallAirway2
LargeAirway2
Trachea
Macrophage2

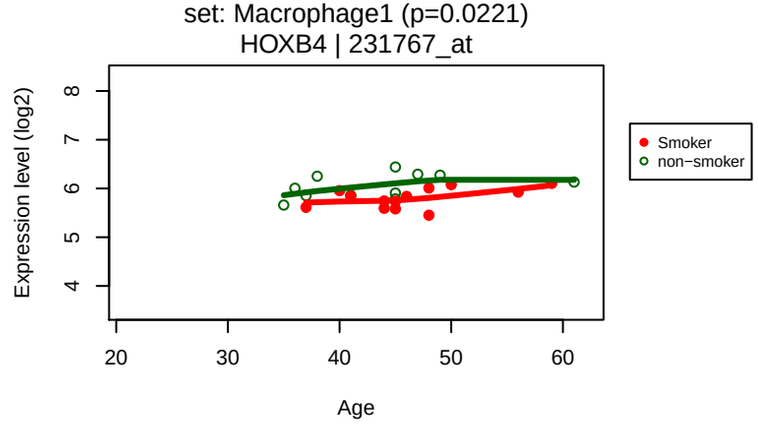

set: SmallAirway1 (p=0.0803)
HOXB4 | 231767_at
Expression level (log2)
Age
Smoker
non−smoker

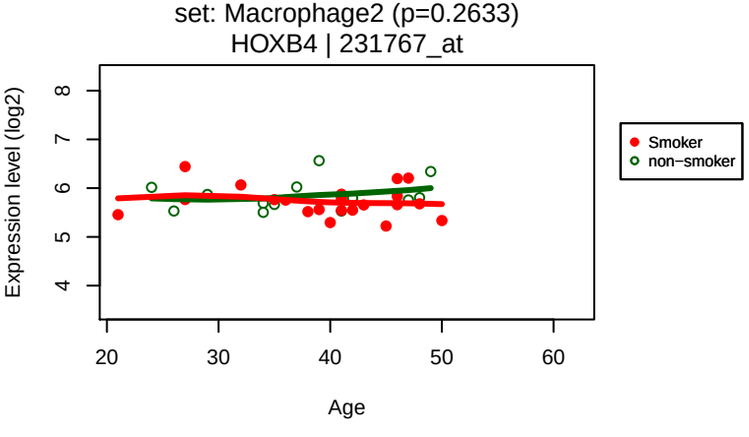

set: SmallAirway2 (p=0.1709)
HOXB4 | 231767_at
Expression level (log2)
Age
Smoker
non−smoker

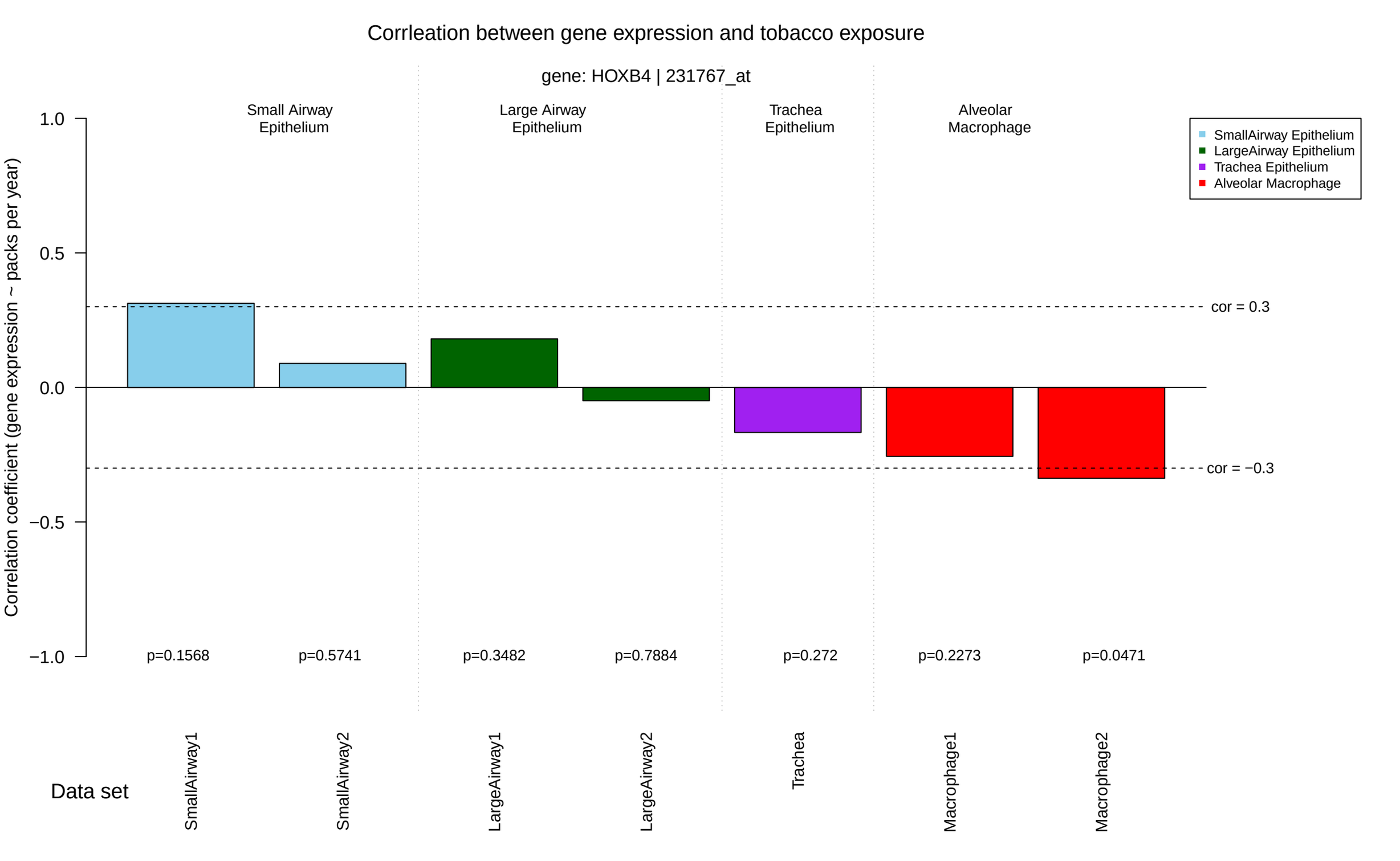

set: LargeAirway1 (p=0.019)
HOXB4 | 231767_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7113)
HOXB4 | 231767_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7435)
HOXB4 | 231767_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0221)
HOXB4 | 231767_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2633)
HOXB4 | 231767_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HOXB4 | 231767_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1568
p=0.5741
p=0.3482
p=0.7884
p=0.272
p=0.2273
p=0.0471
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2